Diary entry for Spectra

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1 What are spectra?

(Literal question, not philosophical.)

Definition 1.1 (Spectrum)**.** A **spectrum** X is a sequence of based spaces $\{X_n\}_{n\geq 0}$ and a sequence of maps $\xi_n \colon \Sigma X_n \to X_{n+1}$, called **bonding maps**. A morphism of spectra $f \colon X \to Y$ is a sequence of maps $f_n \colon X_n \to Y_n$ with the obvious commutativity property:

$$\begin{tikzcd} \Sigma X_n \arrow[r, "\xi_n"] \arrow[d, "\Sigma f_n"] & X_{n+1} \arrow[d, "f_{n+1}"] \\ \Sigma Y_n \arrow[r, "\eta_n"] & Y_{n+1} \end{tikzcd}$$

(This definition is sometimes used instead for so-called "prespectra". If we need to distinguish these from other kinds of spectra we will call them "sequential spectra".)

Note that we will always work with based spaces here unless otherwise specified.

Example 1.2 (Zero spectrum)**.** The zero spectrum $*$ is the spectrum whose maps spaces are all the singleton spaces, which we will also denote as $*$. Since $\Sigma(*) = *$, there is exactly one choice for the bonding maps. The zero map between two spectra X and Y is the unique map factoring through $*$, i.e. sending each X_n to the basepoint of Y_n.

Example 1.3 (Suspension spectra)**.** The sphere spectrum $\mathbb{S}$ is the spectrum whose nth space is the sphere S^n and whose bonding maps $\xi_n \colon \Sigma S^n \to S^{n+1}$ are each the identity map, noting that $\Sigma S^n = S^n \wedge S^1 \cong S^{n+1}$. More generally, for a space X, the suspension spectrum $\Sigma^\infty X$ is the spectrum whose nth space is $\Sigma^n X$ and the bonding maps $\Sigma\Sigma^n X = \Sigma^{n+1} X \to \Sigma^{n+1} X$ are the identity maps.

Definition 1.4 (Stable homotopy groups)**.** For a spectrum X, the kth **stable homotopy group** of X, denoted $\pi_k(X)$ for an integer k, is the colimit of the system of abelian groups

$$\cdots \longrightarrow \pi_{n+k}(X_n) \xrightarrow{\sigma} \pi_{n+1+k}(\Sigma X_n) \xrightarrow{\xi_n} \pi_{n+1+k}(X_{n+1}) \to \cdots$$

where σ is the map sending $\varphi \colon S^{n+k} \to X_n$ to the map $\sigma(\varphi) \colon S^{n+k+1} \to \Sigma X_n$ defined by $\sigma(\varphi)(x,t) = \varphi(x)$. (If k is negative then the system is defined starting at $n \geq -k$).

To emphasize the last sentence: we can have *negative dimensional homotopy groups*! In some sense, spectra let us think about "negative dimensional geometry".

Example 1.5 (Shifted spectra)**.** The colimit defining the k-th stable homotopy group does not change if one deletes finitely many terms (say the first N) from the directed system. Therefore, shifting a spectrum up or down (filling in with singleton spaces or truncating the spectrum respectively to account for indexing) just shifts the homotopy groups. In light of this, for a space X and any integer d, define the spectrum $\Sigma^{\infty-d} X$ to be the suspension spectrum shifted up d spaces (so down $|d|$ spaces if $d < 0$), i.e. $(\Sigma^{\infty-d} X)_n = (\Sigma X)_{n-d}$, where $(\Sigma X)_k = *$ if $k < 0$. The homotopy groups of $\Sigma^{\infty-d} A$ are $\pi_k(\Sigma^{\infty-d} A) = \pi_{k-d}(\Sigma^\infty A)$.

The d-sphere $\mathbb{S}^d$ is then defined as $\Sigma^{\infty-d} S^0$, i.e. the spectrum which, when suspended d times, is the usual sphere spectrum of Example 1.3.

A **degree k element** of a spectrum X is a map $\varphi \colon S^{n+k} \to X_n$, up to identification of φ with $\sigma(\varphi)$, and the stable homotopy groups are the degree k elements of X up to homotopy. what does this identification mean

Definition 1.6 (Equivalences of spectra)**.** A **stable equivalence** of spectra is a map of spectra inducing isomorphisms on every homotopy groups. Two spectra X and Y are **stably equivalent** if there is a sequence of spectra $X = X_1, X_2, \ldots, X_{2n} = Y$ and stable equivalences $X_1 \to X_2$, $X_3 \to X_2$, $X_3 \to X_4$, $\ldots$, $X_{2n-1} \to X_{2n}$ (the parity doesn't change anything since one can pad out the sequence with copies of Y). One says that X and Y are connected by a zig-zag of stable equivalences.

A **level equivalences** is a map of spectra $X \to Y$ in which each map $X_n \to Y_n$ is a weak equivalence.

A **homotopy equivalence** is a map of spectra $f\colon X \to Y$ such that there is a map $g\colon Y \to X$ with the property that $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$ (we will define homotopy later).

We can now define stable homotopy theory: **stable homotopy theory** is the study of spectra up to stable equivalence.

Example 1.7 (Inclusion as stable equivalence)**.** For a spectrum X, let X' be the spectrum formed by replacing the first d spaces with singleton sets. The inclusion $X' \hookrightarrow X$ (i.e. the map on each level is the inclusion) is a stable equivalence.

Example 1.8 (Weak contractibility)**.** For any spectrum X there are unique maps $* \to X \to *$; if one of these (equivalently both) are stably equivalences, then X is said to be weakly contractible. Note that this depends not only on the spaces in the spectrum but also on the bonding maps; for example, any spectrum can be made weakly contractible by setting the bonding maps to the maps sending ΣX_n to the basepoint of X_{n+1}.

Example 1.9 (Thom spectra)**.** Recall that the **Thom space** $T(E)$ of a (real) rank n vector bundle $p\colon E \to B$ is formed by compactifying each fiber to create an n-sphere bundle and then identifying all the points at infinity. Equivalently, $T(E)$ is homeomorphic to $D(E)/S(E)$, where, for a given metric on the vector bundle, $D(E)$ is the unit disk bundle and $S(E) = \partial D(E)$ is the unit sphere bundle. For example, the Thom space of a trivial bundle $p\colon \mathbb{R}^n \times B \to B$ is homeomorphic to $B_+ \wedge S^n = \Sigma^n_+ B$, where B_+ is the space $B \cup \{*\}$ with basepoint $*$ (thus in general the Thom space is a "twisted suspension" of the base).

Letting ε^q be the trivial rank q vector bundle over B and E any other vector bundle, observe that $T(E + \varepsilon^q) = T(E \times \mathbb{R}^q) = \Sigma^q T(E)$. prove vb fact

For another vector bundle E', let E'' be such that $E' + E'' = \varepsilon^q$, and form the formal difference $E - E' = E + E'' - E'' - E' = E + E'' - \varepsilon^q$.